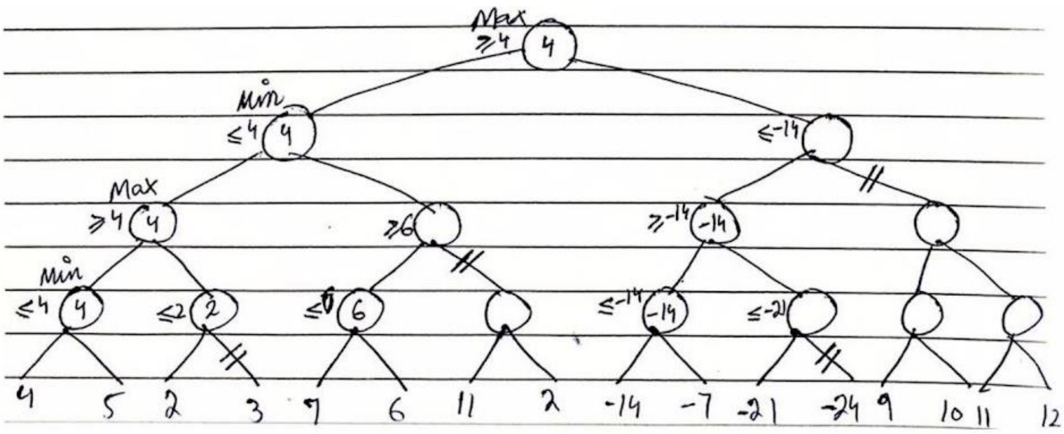
AI-2002

Quiz # 2 (with Solution)

1. A machine learning model is developed to detect whether an email is Spam or Not Spam.
   After testing, the model produced the following results:
   It correctly identified 40 spam emails as spam.
   It incorrectly labelled 10 spam emails as not spam.
   It incorrectly labelled 5 non-spam emails as spam.
   It correctly identified 45 non-spam emails as not spam.
   Based on this information, what are the Accuracy, Precision, and Recall of the model (consider Spam as the positive class)?
   A. Accuracy = 85%, Precision = 88.89%, Recall = 80%
   B. Accuracy = 80%, Precision = 85%, Recall = 88.89%
   C. Accuracy = 90%, Precision = 80%, Recall = 88.89%
   D. Accuracy = 85%, Precision = 80%, Recall = 90%

2. A two-player game is played between MAX and MIN using the Minimax algorithm with Alpha-Beta pruning. MAX is the first player, and the branching factor is **2.**
   The terminal values (from left to right) are:
   4, 5, 2, 3, 7, 6, 11, 2, -14, -7, -21, -24, 9, 10, 11, 12
   You are required to construct the game tree, apply Minimax with Alpha-Beta pruning, and compute all intermediate backed-up values (on your page).
   Based on your solution, which of the following options is correct?
   A. Root value = 4, Best move = Left subtree, Pruned nodes = (3, -24, 9, 10, 11, 12)
   B. Root value = 4, Best move = Left subtree, Pruned nodes = (3, 11, 2, -24, 9, 10, 11, 12)
   C. Root value = 5, Best move = Left subtree, Pruned nodes = (-21, -24, 9, 10)
   D. Root value = -14, Best move = Right subtree, Pruned nodes = (7, 6, 11, 2)



3. A financial institution is developing a system to predict whether a person will default on a loan based on the following categorical attributes:
   Employed (Yes / No);   Credit History (Good / Bad);   Owns House (Yes / No)

A dataset of past applicants is provided in the table below.
Using the Naïve Bayes classifier, predict whether a new applicant with the following profile will default: **Employed: No; Credit History: Bad; Owns House: No**
Students are required to show all probability calculations (prior, likelihood, posterior).

Employed	Yes	Yes	No	Yes	No	No	Yes	Yes	No	No
Credit History	Good	Good	Bad	Bad	Good	Bad	Bad	Good	Good	Bad
Owns House	Yes	No	No	Yes	No	No	No	Yes	Yes	Yes
Default	No	No	Yes	No	Yes	Yes	No	No	No	Yes

A. Default = No, because posterior probability for "No" is higher than "Yes"
B. Default = Yes, because posterior probability for "Yes" is higher than "No"
C. Default = Yes, because prior probability of "Yes" is greater than "No"
D. Default = No, because likelihood of "No" is higher but posterior is lower

Solution:

Step 1: Prior Probabilities for Target

Let N = 10 P(Yes) = 4/10, P(No) = 6/10

For Default

Employment:

Employment	Yes	No
yes	0/4	5/6
No	4/4	1/6

Credit History:

Credit History	Yes	No
Good	1/4	4/6
Bad	3/4	2/6

Own House:

Own House	Yes	No
yes	1/4	4/6
No	3/4	2/6

Classify Given Instance

Given:

- Employment = Unemployed
- Credit History = Bad
- Own House = No

For Default = Yes:

P(D = yes) × P(Unemployed | D = yes) × P(Bad | D = yes) × P(No | D = yes)
= 0.4 × (4/4) × (3/4) × (3/4) = 0.225

For Default = No:

P(D = no) × P(Unemployed | D = no) × P(Bad | D = no) × P(No | D = no)
= 0.6 × (1/6) × (2/6) × (2/6) = 0.01

Conclusion: **0.28 > 0.01 → Classify as Default = YES**

Note:

No Laplace correction needed in this example. It is only required in cases where probabilities are zero.
For example: P(Employed | Default = yes) = (0 + 1) / (4 + 2) = 1/6